
EC 101A: Computer Systems & Programming

Course Code	EC 101A
Course Name	Computer Systems & Programming
L-T-P-C	3-0-2-4
Prerequisites	NIL Sem. Both
One-time approval	5th Senate (Oct 2013)

Course Contents

1. **Basic Computer Fundamentals:** Introduction to computer systems; number system integer, m signed integer, fixed and floating point representations; IEEE standards integer and floating point arithmetic; CPU organization ALU, registers, memory, the idea of program execution at micro level: concept of flow chart and algorithm, algorithms to programs Concept of strain, normal and shear strain, two dimensional state of principal strains, Poisson's ratio, volumetric strain, strain circle.
2. **Basic Programming elements in C++:** Input/Output: Constants, variables expressions and operations: Naming conventions and styles: Conditions and selection statements; Looping and control structures; File I/O header files, string processing; Pre-processor directives such as include, #define, #ifdef, #ifndef, Compiling and linking;
3. **Programming through functional decomposition:** Functions (void and value returning) parameters, scope and lifetime of variables, passing by value, passing by reference passing by constant reference : Design of functions and their interfaces (concept of functional decomposition), recursive functions, function overloading and default arguments; Library functions;
4. **Aggregate data-types:** Arrays and pointers: Structures : Dynamic data and pointers, dynamics arrays ; Introduction to data structures, use of pointers in linked structures;
5. **Object Oriented Programming Concepts:** Data hiding, abstract data types, classes, access control : Class implementation – default constructor, constructors, copy constructor, destructor, operator overloading, friend functions ; Object oriented design, inheritance and composition : Dynamic binding and virtual functions ; Polymorphism ; Dynamic data in classes.

References

1. Dietel H. M. and Dietel P. J., **C ++ How to Program**, Prentice Hall Publications.
2. Nell D., Chip W. and Mark H., **Programming and Problem Solving with C++**, CBS Publishers and Distribution.
3. Cohoon J. P. & Davidson J. W., **C++ Program Design**, Tata McGraw Hill.